Software Requirements Specification (SRS)
College Admission Management System

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1. Introduction

1.1 Purpose

The purpose of this SRS document is to define the comprehensive requirements for the College Admission Management System. This system will facilitate a streamlined, secure, and efficient process for managing prospectus, online applications, merit-list generation, quota-based seat allocation, document verification, and fee collection for educational institutions.

1.2 Scope

The system enables colleges and universities to manage end-to-end admission workflows including online applications, merit calculation, quota-based seat allocation, document verification, and fee collection with role-based access for staff and applicants. Key functionalities include:

- Applicant and staff authentication with role-based access control
- Online multi-step application with document upload and DigiLocker integration
- Merit rank calculation using configurable per-programme formula
- Quota-based seat allocation: General, OBC, SC/ST, EWS, Sports, Defence
- Round-based allotment with upgrade option and waitlist management
- Fee collection with UPI/card payment and scholarship management
- Notification system for application status and allotment results
- Analytics and AISHE/UGC regulatory reporting
- Compliance with national admission and reservation guidelines
- GSTIN-compliant fee receipts and GST invoices

1.3 Definitions, Acronyms, and Abbreviations

- UG: Under-Graduate
- PG: Post-Graduate
- OBC: Other Backward Class
- SC/ST: Scheduled Caste / Scheduled Tribe
- EWS: Economically Weaker Section
- AISHE: All India Survey on Higher Education
- DigiLocker: Government digital document wallet
- JEE/NEET: National entrance examination bodies
- GSTIN: Goods and Services Tax Identification Number
- API: Application Programming Interface

1.4 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- Spring Boot Security Documentation
- React.js Documentation
- AISHE Data Collection Methodology
- UGC Regulations on Admission and Reservation

1.5 Overview

Covers all requirements for the College Admission Management System with role-based security and comprehensive college admission management.

2. Overall Description

2.1 Product Perspective

Spring Boot backend + React.js frontend + JWT authentication + JPA database operations for college admission.

2.2 Product Functions

- Applicant registration with Aadhaar OTP verification
- Online multi-step application with preference list
- Configurable merit formula stored in database per programme
- Quota seat matrix with round-based allotment using SELECT FOR UPDATE
- DigiLocker integration for academic certificate fetch and delivery
- Fee collection with scholarship deductions and GST receipts
- Waitlist management with automatic candidate promotion
- Statistical reports for UGC and AISHE regulatory submissions
- Notification system for merit list, allotment, and deadlines
- Document verification workflow for counsellors and coordinators

2.3 User Classes and Characteristics

- System Administrator: Full access, user management, and compliance oversight
- Applicant: Application submission, status tracking, and fee payment
- Counsellor: Document verification and applicant query resolution
- Programme Coordinator: Merit list generation and seat allocation management
- Registrar: Final admission approval, all reports, and regulatory submissions
- Finance Officer: Fee processing, scholarship management, and GST receipts

2.4 Operating Environment

- Server: Spring Boot 3.x with Spring Security (Java 17+)
- Database: JPA-compatible relational database with college admission management schema
- Frontend: React.js with authentication components (port 8081)
- Backend: Spring Boot with Spring Security REST APIs (port 8080)
- Authentication: JWT-based with role-specific token claims and expiry
- Testing: JUnit 5 for backend, React Testing Library for frontend
- Deployment: Docker containerised on cloud (AWS, Azure, or GCP)

2.5 Design and Implementation Constraints

- Spring Security + JWT on all application and admission endpoints
- Name: alphabetic only; Phone: exactly 10 digits
- Seat allocation uses SELECT FOR UPDATE to prevent concurrent double-allocation
- Merit formula stored in database — not hardcoded — to allow annual changes
- DigiLocker integration required for document fetch and delivery
- Waitlist promotion is transactional to prevent duplicate candidate promotion

2.6 User Documentation

- Applicant online application and status tracking guide
- Counsellor document verification workflow guide
- Coordinator merit list and seat allocation management guide
- Registrar admission closure and regulatory reporting guide

2.7 Assumptions and Dependencies

- DigiLocker API for document fetch from government wallet
- UPI and card payment gateway for fee collection

- Email and SMS services for allotment and status notifications
- AISHE/UGC export format specifications for regulatory submissions

3. Specific Requirements

3.1 External Interfaces

3.1.1 User Interfaces

- Landing Page: Institution overview with prospectus and application login
- Application Form: Multi-step form with preference list and document upload
- Application Status: Timeline tracker with stage-wise status display
- Merit List: Programme-wise rank display with category filters
- Seat Allotment: Round-wise allotment results with upgrade option
- Document Verification: Counsellor checklist with approval interface
- Admin Panel: User management, programme configuration, compliance reports

3.1.2 Hardware Interfaces

- Document scanner for physical document digitisation
- Digital signature pad for contract and form signing
- Barcode/QR scanner for item and record identification

3.1.3 Software Interfaces

- Secured JWT REST API for all protected endpoints
- Email service (SMTP/SendGrid) for notifications and alerts
- SMS gateway for critical alerts and OTP delivery
- Document storage service for file upload, versioning, retrieval

3.1.4 Communication Interfaces

- HTTPS with TLS 1.3 for all client-server communications
- JSON with security headers (CORS, CSP, X-Frame-Options) on all responses
- JWT HMAC-SHA256 signature verification on every API request
- WebSocket for real-time notifications and live dashboard updates

3.2 Functional Requirements

User Authentication and Access Management

FR1: Comprehensive User Registration and Authentication System

Description: The system shall provide admission-specific registration for applicants and staff with identity verification.

Detailed Requirements:

- Registration form captures name, email, phone, role, category, and academic details for all user types
- Applicant registration supports Aadhaar OTP identity verification to prevent duplicate applications
- Staff registration validates employee ID against the institution directory before account activation
- Account activation uses automated email OTP for applicants and manager approval for all staff roles
- Multi-role support handles applicant, counsellor, coordinator, registrar, finance, and admin accounts
- Password security enforces minimum 8 characters with uppercase, lowercase, digit, and special character
- Account recovery uses time-limited email OTP with 15-minute validity for all user types
- Profile management allows post-registration updates to contact details with full audit trail
- Security logging captures every authentication event with IP address, device, and timestamp

- Two-factor authentication optional for standard roles; mandatory for financial operations

FR2: Secure Authentication and Session Management

Description: The system shall implement secure session handling for admission data.

Detailed Requirements:

- JWT tokens: 8 hours for applicants, 12 hours for staff, 24 hours for administrators
- Multi-credential login accepts application number, registered email, or Aadhaar-linked mobile
- Automatic logout after 30 minutes of user inactivity across all sessions
- Device recognition requires additional verification for access from new or unrecognised devices
- Progressive lockout: 15 minutes, 30 minutes, and 24 hours after 5, 10, and 15 failed attempts
- Single sign-on integration with institutional Microsoft 365 for staff convenience
- Emergency access override procedures for critical admission deadline scenarios
- Concurrent session limit enforced per user role to prevent credential sharing
- Token blacklisting on logout, password change, and detected security events
- All session events logged for compliance audit trail and security monitoring

FR3: Role-Based Access Control

Description: The system shall enforce role-appropriate access to all admission data and operations.

Detailed Requirements:

- Applicant role: own application data, status, documents, and payment records only — no access to others
- Counsellor role: document verification functions and applicant query resolution within assigned sessions
- Coordinator role: merit list generation, seat allocation management, and programme configuration access
- Registrar role: final admission approval, all reports, regulatory submissions, and compliance oversight
- Finance role: fee processing, scholarship approvals, GST receipt generation, and payment reporting
- Administrator role: complete access including user management and system configuration
- Seat allocation functions locked between official allocation rounds to prevent interference
- Temporal controls enforce access only during open application and allotment windows
- All access events logged with user identity, timestamp, and operation performed
- No cross-applicant data leakage enforced at the API service layer

Application and Admission Processing

FR4: Online Application and Document Management

Description: The system shall provide a comprehensive multi-step application with DigiLocker integration.

Detailed Requirements:

- Multi-step application captures personal details, academic history, ranked preference list, and documents
- Preference list allows up to 5 programme choices ranked by applicant in order of preference
- Document upload supports marksheets, category certificate, identity proof, and income certificate
- DigiLocker integration fetches academic certificates directly from the government digital wallet
- Application fee payment integrated into the workflow before submission can be confirmed
- Save-and-resume feature allows applicants to complete the form across multiple browser sessions
- Validation enforces name (alphabetic only), phone (exactly 10 digits), and percentage (0–100)

- Duplicate detection prevents a second application using the same Aadhaar number in the same year
- Application acknowledgement email and SMS sent with unique reference number on submission
- Document checklist shows upload completion status for each required item before final submission

FR5: Merit Calculation and Rank Generation

Description: The system shall calculate programme-wise merit ranks using configurable formulas.

Detailed Requirements:

- Merit formula stored as JSON in the Programmes table — not hardcoded in application code
- Formula supports weighted combination of board percentage, entrance score, and extracurricular marks
- Category-specific merit lists generated for General, OBC, SC/ST, EWS, Sports, and Defence quotas
- Configurable tie-breaking rules per programme such as higher maths score for engineering programmes
- Merit list preview available to coordinators for review before official publication to applicants
- Official merit list publication sends simultaneous email and SMS notifications to all applicants
- Historical merit data preserved for each academic year for regulatory reporting and audit
- Merit recalculation on document correction with versioned audit trail of all formula runs
- Applicant rank and category rank both displayed on the applicant portal after publication
- Merit list exported in Excel and PDF formats for institutional records and regulatory submission

FR6: Seat Allocation and Waitlist Management

Description: The system shall perform round-based allocation with quota enforcement and waitlist promotion.

Detailed Requirements:

- Seat matrix configures total seats per programme per quota category per allocation round
- Allocation uses SELECT FOR UPDATE to prevent any concurrent double-allocation race condition
- Round 1 fills first-preference seats; seats unfilled in Round 1 open for Round 2 preferences
- Upgrade option allows allotted candidates to indicate preference for a higher-ranked programme
- Waitlist managed in strict merit rank order with automatic promotion when vacancy occurs
- A candidate may hold only one seat across all programmes at any given time in the system
- Allotment result published simultaneously via portal, email, and SMS to all applicants
- Seat acceptance window: 72 hours after allotment notification before seat is vacated automatically
- Seat vacation triggers immediate waitlist promotion and notification to the next candidate
- Allocation analytics report fill rates, quota utilisation, and waitlist conversion per programme

FR7: Document Verification Workflow

Description: The system shall manage document verification for all admitted candidates.

Detailed Requirements:

- Document checklist auto-generated per programme based on applicant category and quota
- Counsellor marks each document as Verified, Pending, or Discrepant with mandatory remarks
- DigiLocker-fetched documents automatically verified against government records on import
- Discrepancy notification sent to applicant within 2 hours with specific correction instructions
- Conditional admission issued for minor pending documents with a 15-day rectification deadline
- Verification completion triggers generation of the official admission confirmation letter
- Document tamper detection alerts counsellor when a re-uploaded file hash does not match original
- Senior counsellor override resolves verification disputes with documented reason and approval
- Complete verification audit trail records counsellor identity, action, timestamp, and document version
- Post-admission document audit report generated for institutional accreditation submissions

Fee, Analytics, and Administration

FR8: Fee Collection and Scholarship Management

Description: The system shall process admission fees with scholarship support and GST-compliant receipts.

Detailed Requirements:

- Online payment via UPI, card, and net banking through a government-approved payment aggregator
- Scholarship application workflow captures income certificate, merit proof, and category documentation
- Finance Officer approval workflow with configurable multi-level approval for scholarship grants
- Fee structure configurable per programme with tuition, hostel, and examination components
- GST-compliant invoice generated with HSN code and applicable tax rates for every payment
- Fee receipt emailed to applicant within 2 minutes of successful payment gateway confirmation
- Refund workflow processes cancellation refunds within 15 working days of formal request
- Outstanding fee dashboard shows dues per admitted batch and department for Finance team
- Fee analytics report total collections, refunds, scholarship waivers, and outstanding amounts
- Late payment beyond grace period triggers automatic seat cancellation with notification

FR9: Analytics and Regulatory Reporting

Description: The system shall generate admission statistics for institutional planning and regulatory bodies.

Detailed Requirements:

- Application funnel analytics tracks applied, shortlisted, allotted, admitted, and joined counts
- Category-wise and programme-wise statistics with year-over-year comparison charts
- AISHE data export in prescribed format for annual government survey submission
- UGC compliance report with programme-wise intake, actual admissions, and vacancy data
- Scholarship distribution report by category and programme for institutional audit
- Financial summary report: fees collected, refunds processed, scholarships granted, net revenue
- Diversity index report measuring admission representation across all reserved categories
- Predictive demand model uses historical trend data for next admission cycle planning
- Custom report builder for ad-hoc queries by registrar and institutional leadership
- Geographic distribution analysis showing applicant origin by district and state

FR10: Notification and Communication System

Description: The system shall deliver timely admission communications to all stakeholders.

Detailed Requirements:

- Application receipt confirmation via email and SMS immediately after successful submission
- Merit list publication alert dispatched to all applicants simultaneously when results are announced
- Seat allotment result notified via portal, email, and SMS simultaneously to all allotted candidates
- Acceptance deadline reminder sent 24 hours before the acceptance window closes to allotted candidates
- Document discrepancy alert sent within 2 hours of counsellor flagging with correction instructions
- Scholarship award notification with approved amount and terms sent to selected candidates
- Fee receipt emailed within 2 minutes of successful payment gateway confirmation
- Waitlist promotion notification sent immediately when a higher-ranked candidate vacates a seat
- Broadcast announcements dispatched to all active applicants via portal and email
- Notification delivery tracking with failed-delivery automatic retry and escalation

FR11: Exception Handling and Error Management

Description: The system shall implement comprehensive exception handling for all admission operations.

Detailed Requirements:

- Name validation rejects non-alphabetic input with message: 'Name must not contain special characters or numbers'
- Phone validation enforces exactly 10 digits with message: 'Phone Number must be exactly 10 digits long'
- Duplicate application exception prevents second application: 'An application already exists for this Aadhaar'
- Seat conflict exception prevents double-allocation via SELECT FOR UPDATE database transaction
- Payment failure handled with retry guidance and unique reference number for bank follow-up
- Custom exceptions in com.examly.springapp.exception: InvalidNameException, InvalidPhoneException, DuplicateApplicationException, SeatConflictException
- Structured error logs capture applicant context, action, affected record, and timestamp
- User-facing messages are clear and actionable without exposing system internals
- Retry logic for DigiLocker API and payment gateway transient connection failures
- Exception analytics track frequency and peak occurrence by admission stage

FR12: System Administration and Configuration

Description: The system shall provide administrative tools for system and programme management.

Detailed Requirements:

- Programme master management: add, update, and deactivate programmes with seat matrix configuration
- Academic year management defines admission calendar, application window, and allotment round dates
- User lifecycle management: onboard staff, assign roles, and deactivate accounts with full audit trail
- Quota configuration updates reservation percentages per regulatory guideline changes
- Notification template management for all admission event email and SMS content
- System health monitoring tracks API response times, database performance, and error rates
- Daily encrypted backups with weekly restoration testing and secure off-site storage
- Audit log management provides searchable trail of all configuration changes and admin actions
- DigiLocker and payment gateway integration monitoring with status and error dashboard
- Emergency override procedures for extending deadlines and reopening closed application windows

FR13: Mobile and Remote Access

Description: The system shall provide mobile-optimised access for applicants and counsellors.

Detailed Requirements:

- Mobile-responsive web application for complete form completion on smartphones
- Document upload from smartphone camera with automatic compression to meet 5 MB limit
- Offline form completion capability with background synchronisation on reconnect
- Counsellor mobile access for document verification during on-site reporting sessions
- Push notifications for allotment results and deadline reminders to registered mobile
- Mobile UPI QR code payment for fee collection on any device
- Progressive Web App installable from browser without requiring app store download
- Touch-optimised large form fields for comfortable smartphone data entry experience
- Mobile analytics tracking form abandonment rates and mobile versus desktop usage
- Accessibility features for visually impaired applicants including screen reader compatibility

FR14: Data Security and Privacy Protection

Description: The system shall implement comprehensive data protection for all applicant information.

Detailed Requirements:

- AES-256 encryption at rest for sensitive applicant data including Aadhaar and income details
- TLS 1.3 for all data in transit with HSTS headers to prevent protocol downgrade attacks
- Aadhaar numbers stored as one-way SHA-256 hash after identity verification — not retrievable
- Role-based API access controls prevent any cross-applicant data leakage
- PDPB-aligned data deletion procedures available after 7-year mandatory retention period
- Security audit logs record every access to applicant personal data with declared purpose
- Quarterly automated vulnerability scans and annual professional external penetration testing
- All applicant data stored exclusively within Indian data residency jurisdiction
- Signed URLs with 15-minute expiry for all document access preventing unauthorised sharing
- Privacy notice displayed and accepted by applicants at registration before any data collection

FR15: Integration and External System Connectivity

Description: The system shall integrate with government and institutional external systems.

Detailed Requirements:

- DigiLocker API integration for academic certificate fetch from government digital wallet
- NEET and JEE score data import via respective counselling board API for entrance scores
- UPI payment aggregator integration for online admission fee collection
- Aadhaar OTP API for applicant identity verification during registration
- AISHE data export API for automated annual higher education survey submission
- National Scholarship Portal integration for government scholarship application linking
- SIS export of admitted student data to institutional Student Information System
- Email SMTP service for HTML notification emails with PDF attachments
- Audit API for regulatory body secure data access during inspections
- SMS gateway for OTP delivery and admission status notifications

FR16: Compliance and Regulatory Management

Description: The system shall maintain compliance records for regulatory and accreditation requirements.

Detailed Requirements:

- NAAC and NBA data package compiles admission quality metrics in required accreditation format
- UGC compliance report tracks adherence to reservation policy and programme intake norms
- NIRF data export provides programme-wise admission statistics for national ranking submission
- Full audit trail of every seat allocation decision with merit rank and quota basis recorded
- Document retention schedule enforces minimum retention periods per document category
- RTI response package auto-generates requested admission statistics for information requests
- Anti-ragging declaration tracking ensures receipt from all admitted students before joining
- POSH compliance documentation tracks gender-wise admission distribution and committee formation
- Annual compliance calendar generates task list of upcoming regulatory submission deadlines
- Data sharing agreement management tracks institutional data sharing with regulatory bodies

FR17: Advanced Analytics and Predictive Intelligence

Description: The system shall implement predictive analytics for institutional admission planning.

Detailed Requirements:

- Historical trend analysis predicts next cycle application volumes by programme and category
- Category fill rate forecasting helps plan targeted outreach for undersubscribed quota seats

- Dropout prediction model identifies allotted candidates likely not to confirm their seats
- Revenue forecasting models total fee collections based on allotment and historical joining rates
- Geographic distribution analysis identifies applicant origin patterns and travel distance
- Comparative benchmarking against peer institution admission metrics using AISHE data
- Visualisation suite: Sankey diagrams for preference flow and heat maps for geographic origin
- Early warning alerts notify coordinators when fill rates fall below institutional targets per round
- Custom dashboard builder allows registrar to configure KPI views without developer assistance
- Counsellor workload estimation from document verification volumes and verification queue depth

3.3 Performance Requirements

- Authentication: Login < 1 second for 95% of requests
- Record management: Create, read, update, delete < 2 seconds for 90% of requests
- Search: Results returned < 1.5 seconds for complex multi-field queries
- Validation: Combined client and server validation < 1 second for 95% of submissions
- Concurrent users: 500+ simultaneous users without performance degradation
- Reports: Standard reports generate in < 10 seconds for typical data volumes
- Notifications: End-to-end delivery within 5 seconds of trigger event

3.4 Logical Database Requirements

Core Tables:

- Applications Table: id, applicant_id, programme_preferences, entrance_score, board_percentage, status, applied_date
- SeatMatrix Table: id, programme_id, quota, total_seats, allotted, round_no, academic_year
- Programmes Table: id, name, department, merit_formula, intake_capacity, duration_years, is_active
- FeePayments Table: id, applicant_id, amount, payment_mode, transaction_id, paid_at, status

3.5 Design Constraints

- Spring Security + JWT on all protected endpoints
- React with protected routes and mobile-responsive design
- JPA transactions for concurrent data integrity
- All changes logged with user identity and timestamp
- Name: alphabetic only; Phone: exactly 10 digits
- RBAC enforced at both React route and Spring Boot API level

3.6 Software System Attributes

3.6.1 Reliability

- 99.9% uptime with planned maintenance communicated 48 hours ahead
- Graceful non-sensitive error messages for all system failures
- Multi-layer data integrity validation at client, API, service, and database levels
- Automatic failover for critical college admission operations within 60 seconds
- Daily encrypted backups with quarterly restoration testing

3.6.2 Availability

- 24/7 availability with maintenance outside peak operational hours
- Load balancing across minimum two application server instances
- Geographic data replication to secondary data centre for disaster recovery
- Real-time health monitoring with automated on-call escalation
- Horizontal auto-scaling for peak load periods

3.6.3 Security

- Authentication: JWT HS256 with role-specific session duration configuration

- Authorisation: Role-based access control at API service layer for every endpoint
- Data Protection: AES-256 at rest and TLS 1.3 in transit for all sensitive data
- Input Validation: Server-side sanitisation against injection and XSS attacks
- Audit Trail: Immutable logging of all data access and modification events
- Compliance: Regulatory standards verified and documented at every major release

3.6.4 Maintainability

- Layered architecture: Controller, Service, Repository, Domain Model layers
- Minimum 85% unit and integration test coverage measured in CI pipeline
- Swagger/OpenAPI 3.0 documentation with examples for every endpoint
- Git feature branch workflow with mandatory peer code review
- Spring Boot profiles with encrypted secret management

3.6.5 Portability

- Docker + Docker Compose for consistent local and production deployments
- Spring Data JPA: MySQL 8.0+ and PostgreSQL 14+ without code changes
- Cloud manifests: AWS ECS, Azure AKS, GCP GKE
- Spring Boot profiles: dev, test, staging, production
- Cross-browser: Chrome 100+, Firefox 100+, Edge 100+, Safari 15+

3.7 Other Requirements

- Testing: Backend 85% JUnit 5 and Mockito; Frontend 75% React Testing Library
- Security Testing: Quarterly OWASP ZAP + annual professional penetration testing
- Performance Testing: Monthly JMeter load tests before each major release
- Documentation: API reference, deployment runbook, user manuals each release
- Compliance: Education regulatory standards verified at every major release
- Accessibility: WCAG 2.1 Level AA for all user-facing interfaces

4. Appendices

Appendix A: Role-Based Permission Matrix

Functionality	Guest	Applicant	Counsellor	Coordinator	Registrar	Finance	Admin
View Prospectus	✓	✓	✓	✓	✓	✓	✓
Submit Application	✓	✓	-	-	-	-	✓
Track Application Status	-	✓	✓	✓	✓	-	✓
Verify Documents	-	-	✓	✓	✓	-	✓
Generate Merit List	-	-	-	✓	✓	-	✓
Allocate Seats	-	-	-	✓	✓	-	✓
Process Fees	-	-	-	-	-	✓	✓
Generate Reports	-	-	-	L	✓	✓	✓
Manage Programme Config	-	-	-	-	✓	-	✓
Manage Users	-	-	-	-	-	-	✓

Appendix B: Database Schema Details

Applications Table:

- id (PRIMARY KEY, AUTO_INCREMENT)
- applicant_id (FOREIGN KEY → Users.id)
- programme_preferences (JSON) — ranked list of programme IDs
- entrance_score (DECIMAL(6,2))
- board_percentage (DECIMAL(5,2))
- status (ENUM: SUBMITTED, VERIFIED, ALLOTTED, JOINED, CANCELLED)
- applied_date (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)

SeatMatrix Table:

- id (PRIMARY KEY, AUTO_INCREMENT)
- programme_id (FOREIGN KEY → Programmes.id)
- quota (VARCHAR(20)) — GEN/OBC/SC/ST/EWS/SPORTS
- total_seats (INT, NOT NULL)
- allotted (INT DEFAULT 0)
- round_no (INT, NOT NULL)
- academic_year (VARCHAR(10), NOT NULL)

Programmes Table:

- id (PRIMARY KEY, AUTO_INCREMENT)
- name (VARCHAR(200), NOT NULL)
- department (VARCHAR(100))
- merit_formula (JSON) — configurable weightage per component
- intake_capacity (INT, NOT NULL)
- duration_years (INT)
- is_active (BOOLEAN DEFAULT TRUE)

FeePayments Table:

- id (PRIMARY KEY, AUTO_INCREMENT)
- applicant_id (FOREIGN KEY → Users.id)
- amount (DECIMAL(10,2), NOT NULL)
- payment_mode (ENUM: UPI, CARD, NET_BANKING, DD)
- transaction_id (VARCHAR(100), UNIQUE)
- paid_at (TIMESTAMP)
- status (ENUM: PENDING, COMPLETED, FAILED, REFUNDED)

Appendix C: Validation Rules Summary

Field	Validation Rules	Error Messages
Name	Required, alphabetic characters and spaces only, 2–100 characters	"Name is required" / "Name must not contain numbers or special characters"
Phone Number	Required, exactly 10 consecutive numeric digits	"Phone Number is required" / "Phone Number must be exactly 10 digits long"

Email	Required, valid email format, unique per account	"Please enter a valid email address" / "This email is already registered"
Password	Minimum 8 characters with uppercase, lowercase, digit, and special character	"Password must meet security requirements"
Board Percentage	0–100 numeric value	"Board percentage must be between 0 and 100"
Entrance Score	Non-negative numeric value	"Entrance score must be a non-negative number"
Preference Rank	Integer between 1 and 5	"Preference rank must be between 1 and 5"

Appendix D: Security Implementation Details

JWT Configuration:

- Token Expiration: 8 hours for regular users, 12 hours for staff, 24 hours for admins
- Refresh Token: 30-day validity with automatic rotation on each use
- Algorithm: HS256 with minimum 256-bit secret key stored as environment variable
- Secret Key: Never hardcoded in source code — loaded from secure environment config
- Token Blacklisting: Implemented for logout, password change, and security breaches
- Token Claims: Include userId, role, email, issuedAt, and expiration timestamp

Password Policy:

- Minimum 8 characters with uppercase, lowercase, digit, and special character
- Cannot match last 5 passwords; expires 90 days staff / 180 days users
- Lockout 15 minutes after 5 consecutive failed login attempts
- Secure password reset via email OTP with 15-minute validity window
- Password strength indicator displayed in real time during registration

Data Security:

- Sensitive college admission data encrypted at rest using AES-256
- All communications use HTTPS with TLS 1.3 and HSTS headers
- Server-side input sanitisation preventing SQL injection and XSS
- Role-scoped data access controls enforced at service layer
- Tamper-evident encrypted audit logs for all security events

Rate Limiting:

- Login: maximum 5 attempts per 15-minute window per IP address
- Registration: maximum 3 new accounts per day per IP
- Search: maximum 1,000 queries per hour per authenticated user
- Data writes: maximum 100 operations per hour per user
- Password reset: maximum 3 emails per hour per address

Appendix E: Exception Handling Classes

The following custom exception classes shall be implemented in com.examly.springapp.exception:

InvalidNameException

Thrown when a college admission name field contains non-alphabetic characters or digits.

- Extends: RuntimeException
- Trigger: Non-alphabetic characters in name field
- HTTP Status: 400 Bad Request

InvalidPhoneException

Thrown when a phone number is not exactly 10 consecutive digits.

- Extends: RuntimeException
- Trigger: Phone not exactly 10 digits
- HTTP Status: 400 Bad Request

DuplicateApplicantException

Thrown when a college admission record with the same unique identifier already exists.

- Extends: RuntimeException
- Trigger: Duplicate unique identifier on create
- HTTP Status: 409 Conflict

UnauthorisedAccessException

Thrown when a user accesses a resource outside their assigned role.

- Extends: RuntimeException
- Trigger: API call beyond user role permissions
- HTTP Status: 403 Forbidden

ResourceNotFoundException

Thrown when a college admission record cannot be found by the provided ID.

- Extends: RuntimeException
- Trigger: GET/PUT/DELETE with non-existent ID
- HTTP Status: 404 Not Found

Appendix F: Business Rules and Constraints

Application Rules:

- One application per applicant per academic year — enforced via Aadhaar duplicate check
- All required fields and documents must be uploaded before submission is accepted
- Application reference number generated and sent by SMS and email on submission
- Applications cannot be modified after the application window closing date
- Document uploads must include marksheets, identity proof, and category certificate

Seat Allocation Rules:

- Merit formula for each programme stored in the Programmes table — not hardcoded
- SELECT FOR UPDATE prevents concurrent double-allocation across parallel sessions
- Waitlisted candidates promoted automatically when allotted candidates cancel seats
- Reservation quotas strictly enforced per programme per round as configured
- A candidate may hold only one allotted seat across all programmes at any time

Fee and Scholarship Rules:

- Admission fee must be paid within 72 hours of allotment to retain the seat
- Scholarship deductions applied before student pays the remaining balance amount

- Refunds processed within 15 working days of formal cancellation request
- GST-compliant receipts generated for all fee payments
- Late payment beyond grace period triggers automatic seat cancellation

Data Retention Rules:

- Admission records retained for 10 years after academic year closure
- Payment records retained for 7 years for financial audit purposes
- Uploaded documents retained for 5 years after admission closure
- Audit logs retained for 10 years for regulatory compliance and inspection

Appendix G: Future Enhancement Roadmap

Phase 2 Features:

- Native iOS and Android apps with offline college admission capability
- Advanced analytics with machine learning trend forecasting
- Third-party platform integration via REST and webhook APIs
- Automated recommendation algorithms reducing manual work
- Multi-language support for diverse user populations

Phase 3 Features:

- AI for automated pre-screening and intelligent workflow routing
- Predictive analytics for outcome forecasting
- Blockchain-based immutable record management
- Real-time collaborative workspaces with co-editing

Phase 4 Features:

- Continuous ML retraining pipelines from production data
- Self-service BI with drag-and-drop report builder
- AI-assisted data extraction reducing manual entry by 80%
- Open partner API marketplace for third-party extensions

Appendix H: API Endpoint Summary

Application Management:

- POST /api/applications — Submit new application
- GET /api/applications/{id} — Get application details
- PUT /api/applications/{id}/documents — Upload or update documents
- GET /api/applications/merit-list/{progId} — Get programme merit list
- GET /api/applications/status/{id} — Get current application status

Seat Allocation:

- POST /api/allocation/run — Run seat allocation for a round
- GET /api/allocation/results/{round} — Get allotment results
- PUT /api/allocation/{id}/upgrade — Request seat upgrade
- GET /api/allocation/waitlist/{progId} — Get waitlist
- POST /api/allocation/confirm/{id} — Confirm seat acceptance

Fee Management:

- POST /api/fees/initiate — Initiate fee payment
- GET /api/fees/receipt/{id} — Download payment receipt
- POST /api/scholarships/apply — Submit scholarship application
- GET /api/fees/outstanding/{id} — Outstanding fee balance

User Management:

- POST /api/auth/register — Register applicant or staff
- POST /api/auth/login — Authenticate and receive JWT
- POST /api/auth/logout — Invalidate JWT token
- GET /api/users/profile — Get current user profile

Appendix I: Component and Navigation Structure

React Components:

App.jsx:

- React Router DOM v6 with nested and protected route support
- Routes: "/" (Home), "/login" (Login), "/apply" (ApplicationForm), "/status" (ApplicationStatus), "/dashboard" (Dashboard)
- NavBar and Footer in root layout for all pages
- ProtectedRoute redirects unauthenticated users to login
- Global ErrorBoundary for unhandled runtime exceptions

NavBar.jsx:

- Application title: "College Admission Management System"
- Navigation links: "Home" (/), "Login" (/login), "Apply" (/apply), "Status" (/status), "Dashboard" (/dashboard)
- Logged-in user name and role from JWT token in AuthContext
- Logout clears JWT from localStorage and redirects to login
- CSS classes: nav-link, nav-active, nav-brand

Login.jsx:

- Email/username and password with show/hide toggle
- Submit "Login" → POST /api/auth/login
- JWT stored in localStorage; role saved in AuthContext on success
- Redirects to role-specific dashboard from JWT role claim
- Error: "Invalid credentials. Please check your email and password."

Register.jsx:

- Multi-step form: name, email, phone, role, domain-specific fields
- Name: alphabetic + spaces only; Phone: exactly 10 digits
- Email uniqueness verified via debounced API call
- Submit "Register" → POST /api/auth/register
- Success: "Registration successful! Please verify your email."

Dashboard.jsx:

- Role-aware cards, panels, and data tables per authenticated role

- KPI summary cards for most relevant metrics per role
- Recent activity feed with links to detail pages
- Quick-action buttons for three most-used operations
- Responsive CSS grid: single-column mobile, three-column desktop

Footer.jsx:

- "© 2024 College Admission Management System. All rights reserved."
- Rendered via App.jsx layout — not imported individually
- Links to privacy policy, terms of service, and support

Document Version Control

Version	Changes	Author
1.0	Initial SRS document creation for College Admission Management System	Development Team